

HEALTH SCIENCES I NOTES

By:

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BASICS

Positions:

Supine - laying face down
Prone - laying face up
Lateral - away from midline
Medial - toward midline
Proximal - toward attachment point
Distal - away from attachment point
Cranial - toward brain
Posterior - from the back
Dorsal - the back
Anterior - from the front
Ventral - the front
Caudal - toward the butt
Sagittal Plane - vertical (cuts left and right)
Coronal Plane - Horizontal (cut front and back)
Transverse Plane - Parallel to floor (Cut front and back)
Dorsal Cavity - back (head and spine)
Ventral Cavity - front (thoracic and abdominal)
Deep - Inside the body
Superficial - On the surface

Terms

Anatomy - Structure

Gross - seen with naked eye
Microscopic - seen with microscope
Developmental - changes throughout time
Comparative - Human v Animal
Systematic - By system
Embryology - of the embryos

Physiology - Function

Cavities

Abdominal

4 Quadrants, 9 Regions

1 2 3 → Right hypochondriac, Epigastric, Left Hypochondriac
4 5 6 → Right Lumbar, Umbilical, Left Lumbar
7 8 9 → Right Iliac Region, Hypogastric, Left Iliac Region

Necessary activities that allow organisms to maintain homeostasis

- Movement (muscles)
- Ingestion & Digestion & Synthesis & Assimilation (digestive)
- Transport (Circulatory)
- Respiration (Respiratory)
- Growth (Skeletal)
- Secretion (Endocrine)
- Excretion (Urinary)
- Regulation/Decision/Sensation (Nervous)
- Reproduction (Reproductive)

Metabolism

2 Processes

Anabolism - small to big

Catabolism - big to small (carnibal eating)

NEEDED TO MAINTAIN HOMEOSTASIS

Cells

Parts and Functions

- **Cell Plasma Membrane** - selective and semi permeable (outer wall)
- **Nucleus** - control center of the cell contains nuclei and DNA
- **Cytoplasm** - semifluid mechanism btw the membrane and nucleus
- Centrosomes - active during mitosis
- Ribosomes - protein synthesis
- Lysosomes - digest proteins
- Peroxisomes - Digest fats & detoxify
- Mitochondria - Make ATP
- Endoplasmic Reticulum - transport in and out of the nucleus
- Cilia&Flagella - Movement
- Cytoskeleton - Microtubules and internal skeleton of the cell

Reproduction

Mitosis -> DNA distributed to 2 new daughter cells, 46 chromosomes each

Meiosis -> special type of cell division, (ova and sperm), each cell combines to make 46 chromosomes total

Internal Movement

Passive (No energy) ->

Diffusion (greater concentration area to lower concentration area to equalize)

Osmosis (Water occurs across a semi permeable membrane from greater concentration area to lower concentration area to equalize)

Filtration (solutes and water move and get separated)

Active (Energy) ->

Active Transport (Requires ATP to move from greater concentration area to lower concentration area to equalize)

Phagocytosis (Cell eats other cell)

Pinocytosis (Eats other cell in a solution, cell drinking)

TUMORS

Benign → confined to a specific area

Malignant → Cancer causing

TYPES of Tissue

Connective:

- Adipose (loose tissue), Tendons (dense tissue), Embryonic (in the embryo), Collagen (hair and stuff), Fluid Connective (blood and lymph)
- Establishes frameworks, and transports fluids and materials, protects organs, connects other tissues, stores energy (adipose fat)

Epithelial:

- Covers exposed passage ways, internal stuff so organs don't scratch, external skin stuff, makes glands
- Made of only closely packed cells, attached to underlying tissues, lacks blood vessels, get nutrients by diffusion and absorption, damaged cells are replaced with cell divisions,
- Controls permeability and provides secretions and sensations

Muscle

- look muscle unit

Nervous Tissue

- Neurons and brain cells

Organization

Subatomic Particles → Atom → Organell → Cell → Tissue → Organ → Organ System → Organism

TERMINOLOGY

Metric System To Imperial System:

1g=1000mg
1kg=1000g
1L=1000ml
1ml=1cm³
1cm=10mm
100cm=1m
1km=1000m
1ml=15mimim=15drops
5ml=1teaspoon
15ml=1tablespoon
30ml=1oz=2tablespoon
500ml=1pint
1L=1qt
1L water is 1kg
1kg=2.2lbs
1g=1000mg
1g=15grains
1inch=2.5cm
 $C = (F - 32) * (5/9)$
 $F = (C * (9/5)) + 32$

LIST:

Prefixes

1. a / an – without / not
2. Ab – away from
3. Ad – toward
4. Ante – before
5. Anti – against
6. Auto – self

7. Bi – two
8. Brady – slow
9. Contra – against
10. Dys – bad, difficult, painful
11. Ecto – outside
12. Endo – inside
13. Epi – on, upon
14. Ex – out, away from
15. Hemi – half
16. Hyper – above, excessive
17. Hypo – below, deficient
18. Inter – between
19. Intra – within
20. Later – side
21. Leuko – white
22. Lipo – fat
23. Mal – bad / poor
24. Micro – small
25. Neo – new
26. Ortho – straight, correct
27. Peri – around
28. Poly – many

- 29. Post – after
 - 30. Pre – before
 - 31. Pro – forward, in front of
 - 32. Semi – half / partial
 - 33. Sub – under
 - 34. Tach (Tachy-) – fast
-

Suffixes

- 35. -itis – inflammation
- 36. -emia – blood condition
- 37. -oma – tumor / swelling
- 38. -osis – abnormal condition
- 39. -pathy – disease / disorder
- 40. -iasis – pathological/parasite condition
- 41. -algia – pain
- 42. -dynia – pain
- 43. -cele – hernia / protrusion
- 44. -lysis – breakdown / destruction
- 45. -malacia – softening of tissue
- 46. -megaly – enlargement
- 47. -phobia – fear

48. -plegia – paralysis
49. -rrhea – discharge / flow
50. -rrhage / -rrhagia – bursting forth, heavy flow
51. -pnea – breathing
52. -phasia – speech
53. -uria – urine condition
54. -phagia – eating / swallowing
55. -ectomy – surgical removal
56. -otomy – incision
57. -ostomy – creation of an opening
58. -plasty – surgical repair
59. -scopy – visual exam with scope
60. -graphy – process of recording
61. -gram – record or image
62. -therapy – treatment
63. -centesis – puncture to remove fluid
64. -logy – study of
65. -ologist / -logist – specialist
66. -iatry – medical treatment
67. -iatrics – branch of medicine

EMT

Law Enforcement

A critical role in maintaining order and providing security during emergencies, ensure safety of people and manage traffic and crowds, ensure safety of other emergency responders

Firefighters

Most common, usually first on scene, responding to accidents, rescues, medical calls and hazardous materials

Emergency Medical

EMR/EMT/AEMT/Paramedic/CCP/FPC/Community Paramedic

Dispatch

The FIRST first responder

Gather information and share with other responders

Others

Military

Nurses

Correction Officers

All

Selfless

Put themselves in harms way

Miss important days, birthdays and holidays

Ready at a moments notice

Go to training always

Unpredictable work schedule

Early Recognition

Helps shorten recovery time

Helps prevent long term disabilities

Provide comfort to victims

Helps prevent situation

Scene Safety

First thing to check when doing first aid

YOU are the most important person on scene

Initial Assessment

- 1) General Condition & Level of Consciousness (LOC)
- 2) Airway
- 3) Breathing
- 4) Circulation

Good Samaritan Law

Legal Protection to people who give reasonable assistance to someone who is ill or in danger,

If someone is conscious -> Need consent

If unconscious -> Consent Implied

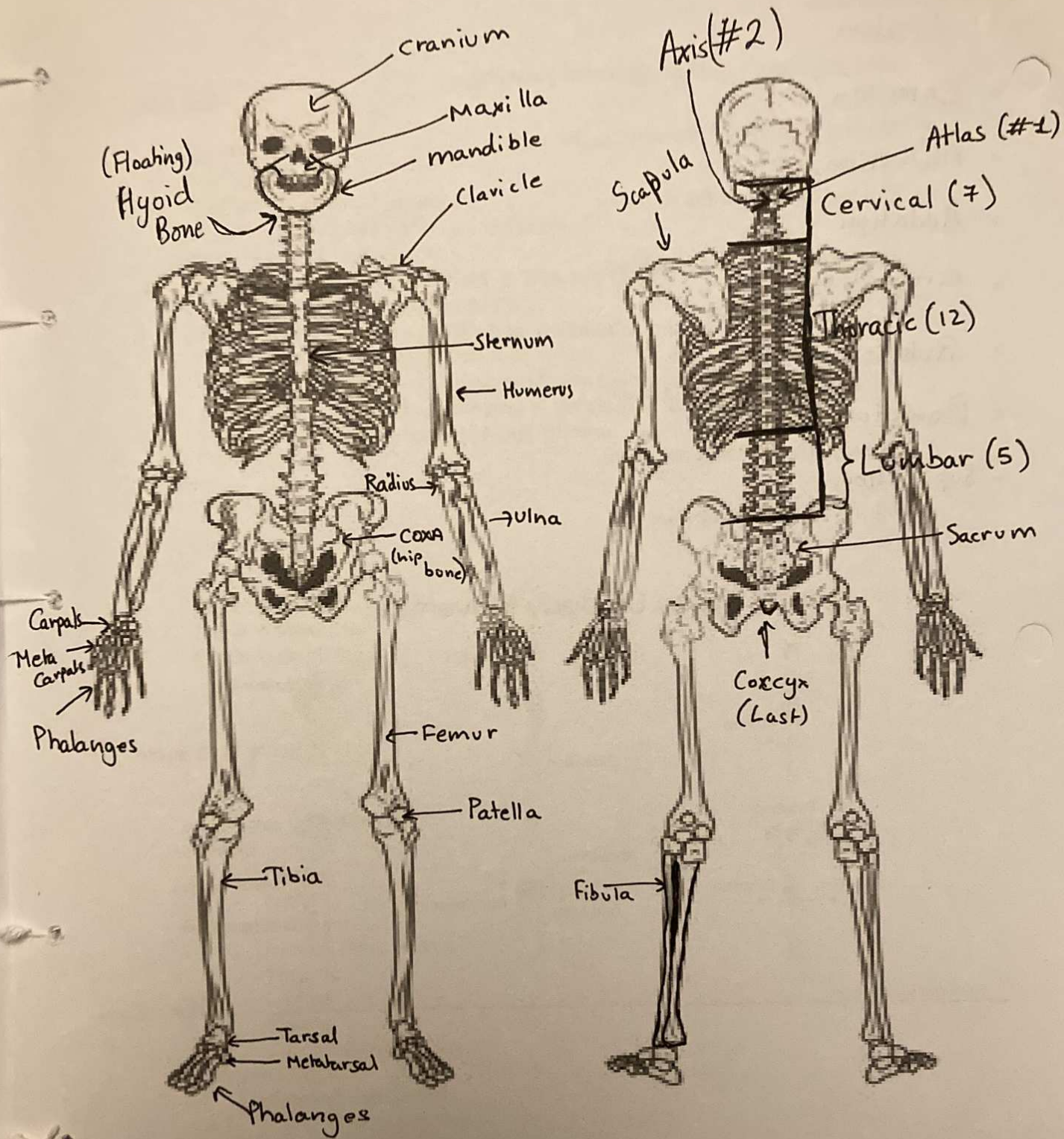
SKELETAL SYSTEM

Functions

- Supports
- Protects
- Allows for movement and attaches muscles
- Mineral Storage
- Make blood

Classifications

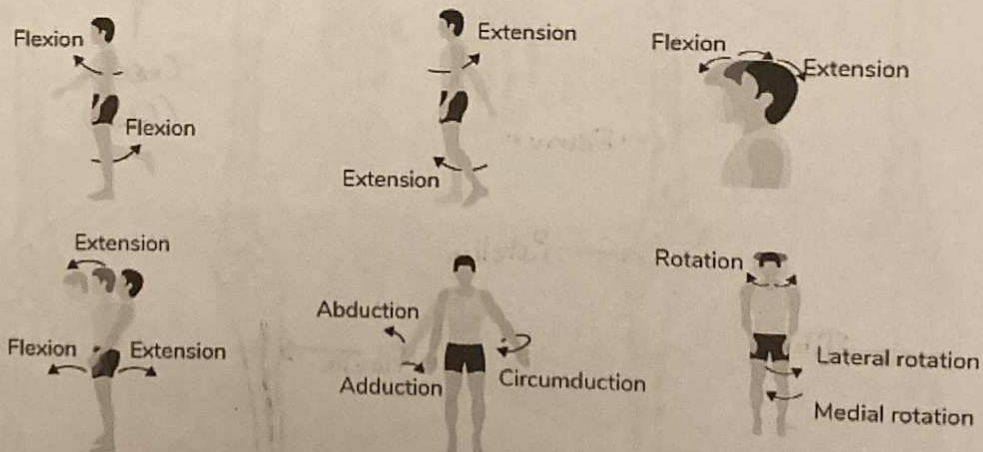
- Long bone
- Short Bone
- Flat Bone
- Irregular Bones



TYPES OF MOTION

- Flexion
 - Bringing closer together, Decreasing the angle
- Extension
 - Moving further part; increasing the angle
- Abduction
 - Moving away from the midline
- Adduction
 - Movement toward the midline
- Circumduction
 - Includes flexion, extension, abduction, and adduction
- Rotation
 - Move around on a central axis
- Pronation
 - Forearm turns hand downward
- Supination
 - Palm is forward or upward

Types Of Body Movement



LONG BONE EXPLAINED

- Epiphysis <- Tip
- Shaft: Diaphysis
- Cartilage covers the epiphysis
- Periostenum <- Tissue that covers the bone
- Sponge Bone <- inside the tip
- Hollow in the shaft contains marrow <- Medullary Cavity
- Endotenum <- covers medullary cavity
- BONE MARROW
 - Makes three types of blood cells
 - Red blood cells (erythrocytes) <- oxygen carry
 - Platelets (thrombocytes) <- clot areas to stop bleed
 - White (leukocytes) <- fight disease
 - Yellow

CELLS

- Osteocytes <- Bone cells
- Blasts <- Make bone
- Clasts <- Break Bone

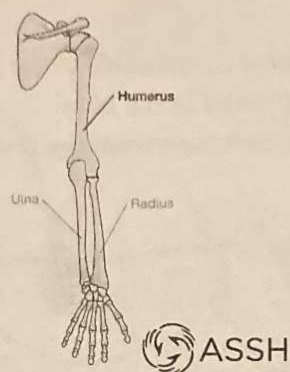
AXIAL SKELETON

- 80
- Skull, ear, vertebral, thoracic
- **Skull**
 - Cranium
 - Facial
- **Vertebre Column**
 - Cervical (Atlas on top)
 - Thoracic
 - Lumbar
 - Sacral
 - Coccygeal (Axis on bottom)
- **Thoracic Cage**
 - Sternum (breastbone)
 - Manubrium (top)
 - Body
 - Xiphoid (bottom)
 - 12 (24) total ribs

- 7 pairs (14) true ribs
- 3 pairs (6) false ribs (attached to 7th rib)
- 2 floating ribs (not connected to front connected to vertebre)
- Protects the thoracic and abomen
- Hyoid
 - ONLY FLOATING BONE
 - AT THE BACK OF NECK
 - SUPPORTS TOUNGE
- Ossicles
 - 3 bones in the year
 - Malleus (hammer)
 - Incus (anvil)
 - Stapes (stirrup)

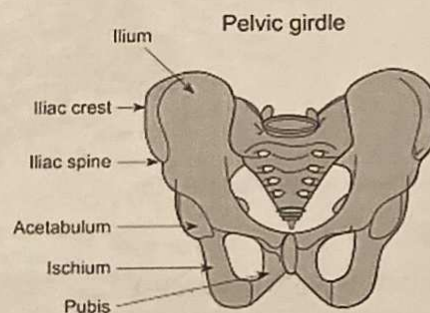
APPENDICULAR SKELETON

- 126
- Pelvis, Limbs
- Pectoral Griddle
 - Clavicle (collar)
 - Scapula (shoulder)
- Forearm
 - Ulna (pinky)
 - Radius (thumb)
- Hand
 - 27 bones
 - Carpals (8, Wrist), Meta carpals (5, palm), Phalanges (14, fingers),
 - 3 per finger, 2 for thumb



PELVIC GIRDLE

- Pelvic girdle consists of 3 bones:
 - Ilium
 - Ischium
 - Pubis
- Fuse together with the sacrum to form the pelvic girdle

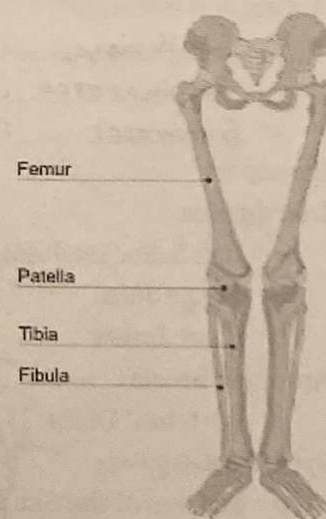


UPPER LEG

- Contains the longest and strongest bone in the body
 - Femur
- Articulates with the acetabulum of the hip and the tibia and fibula

LOWER LEG & KNEE

- Consists of 2 bones:
 - Tibia - largest of the two lower leg bones (medial side)
 - Fibula - lateral side
- Patella - kneecap
 - Flat, triangular shaped, sesamoid bone
 - Ossifies at puberty
 - Surrounded by 4 bursa sacs to help cushion the knee joint

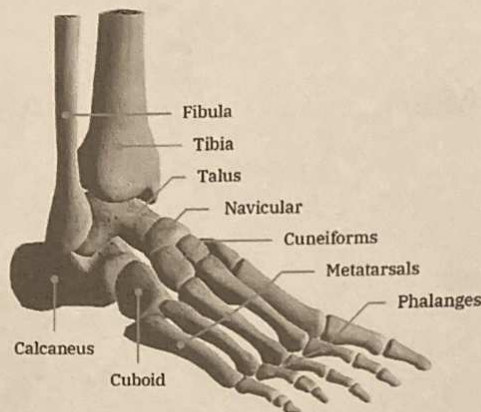


ANKLE & FOOT

- Ankle (tarsus)
 - Contains 7 tarsal bones
 - Largest is the Calcaneus (heel bone)

- Foot

- Contains 5 metatarsal bones - somewhat comparable to the metacarpals of the hand - arranged to form distinct arches not found in the hand
- 14 phalanges in each foot, 3 in each toe except the great toes which only have 2



JOINTS

- Joints or articulations are points of contact between 2 bones
- Three main types, classified according to their degree of movement:
 - Diarthroses (moveable)
 - Amphiaroses (partially moveable)
 - Synaroses (immoveable)

- Examples:

- Diarthroses:

- Ball & Socket Joint
- Hinge Joint
- Pivot Joints

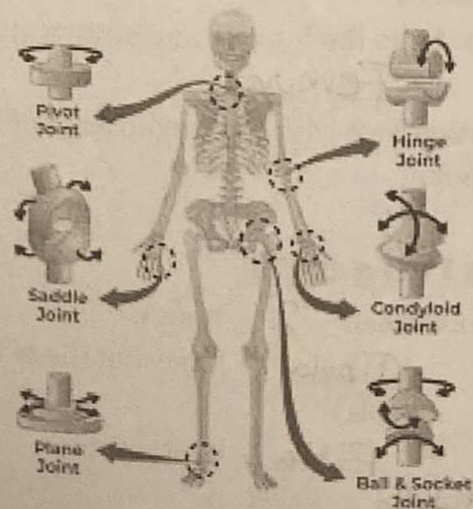
- Amphiarthrosis

- Vertebral Discs

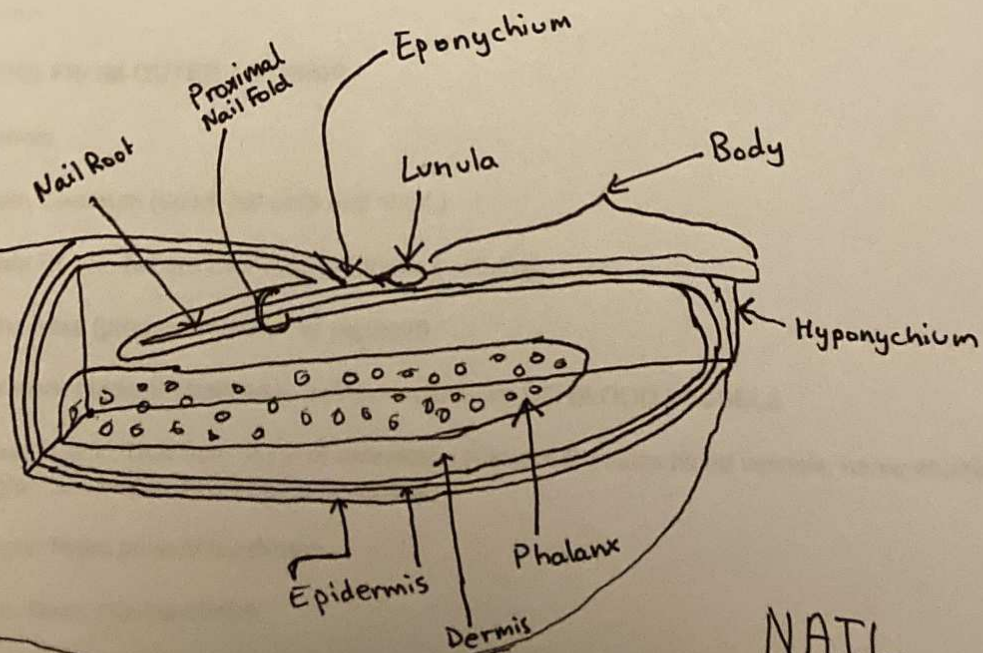
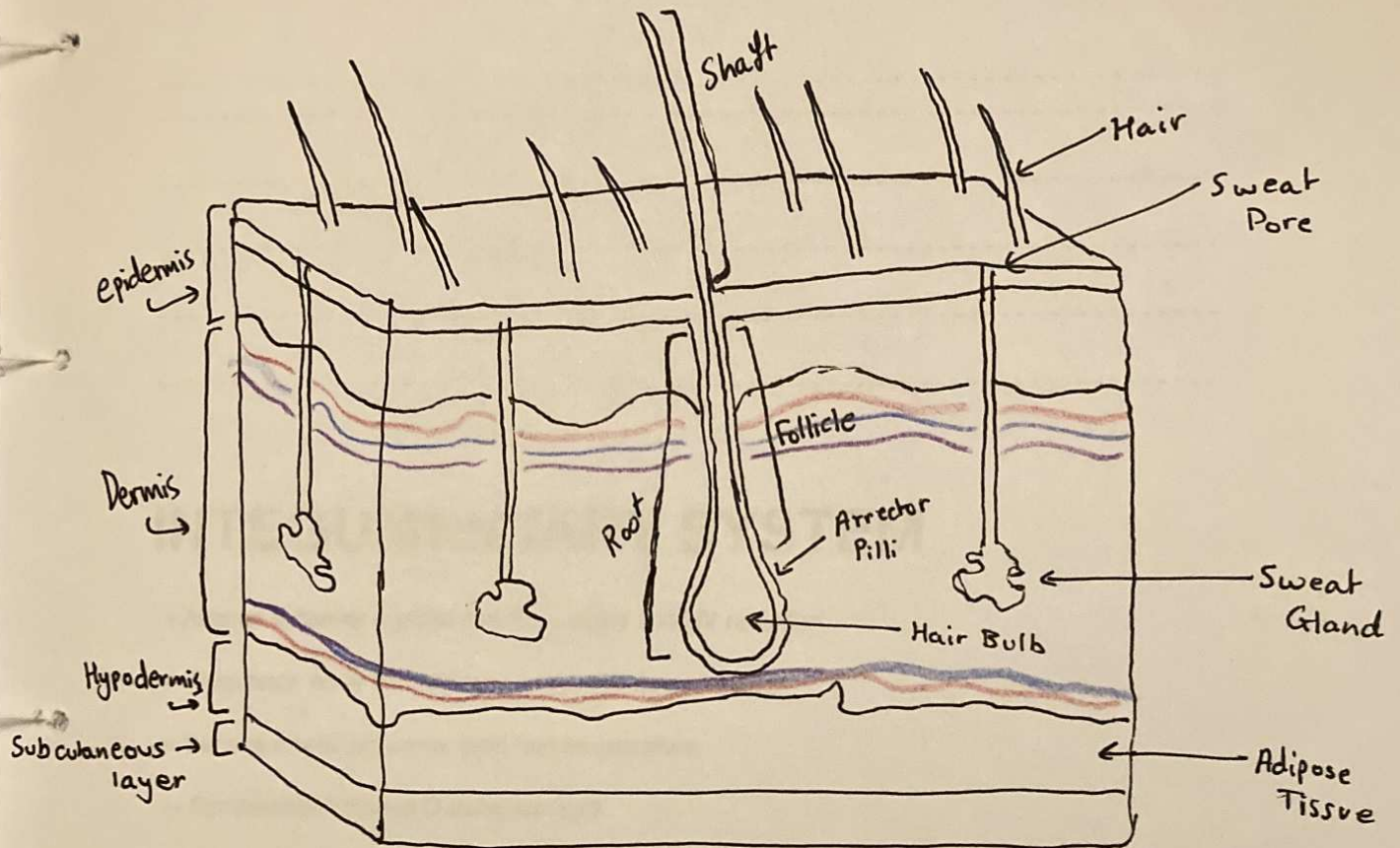
- Synarthrosis:

- Sutures of the Skull

Types of Joints



SKIN



NAIL

INTEGUMENTARY SYSTEM

- Acts as a barrier against infection, injury and UV radiation
- Regulates body temperature and prevents water loss
- Senses touch, pressure, pain and temperature
- Synthesizes Vitamin D using sunlight
- ACTS AS SHOCK ABSORBER ——— Hair Flexible strands of dead, keratinized cells

LAYERS FROM OUTER TO INNER

Epidermis

Stratum Corneum (dead, flat cells that shed)

Stratum Basale (where new cells are made (mitosis))

Melanocytes (produce melanin for pigment)

Outer layer made of stratified squamous epithelium NO BLOOD VESSELS

Dermis (Thick "True Skin" layer of connective tissue) <- Contains blood vessels, nerve endings, and glands Gives skin strength and elasticity

Collagen fibers provide toughness

Elastin fibers provide stretch

Hypodermis (Subcutaneous) Deepest layer, attaches skin to muscle and bone

Adipose Tissue (Fat) -> stores energy and insulates body

Major Blood Vessels

HAIR STUFF

→Hair Follicle -> tubelike depression where hair grows →Root -> part enclosed in follicle

→Arrector Pili -> muscle that causes goosebumps

NAIL STUFF

Protects scalp and filters air (nose/lashes)

Nails Scale-like modifications of the epidermis

Hard keratin sheets protecting tips of fingers and toes

Nail Matrix -> responsible for nail growth

Lunula -> white half-moon at base

Glands

Sebaceous Glands (Oil) Found all over body except palms and soles

Produces Sebum (oil) →Lubricates skin and hair →Kills bacteria →Prevents water loss

Sudoriferous Glands (Sweat) Coiled tubes extending through dermis

Eccrine Glands -> release watery sweat for cooling (thermoregulation)

Apocrine Glands -> release sweat during stress/excitement Sensory Receptors Located in Dermis

Meissner's corpuscles detect light touch

Pacinian corpuscles detect deep pressure Free nerve endings detect pain and temperature

STUFF

Proteins & Pigments → Keratin for Waterproofing/Hardening → Melanin for UV Protection/Color
→ Collagen for Structure

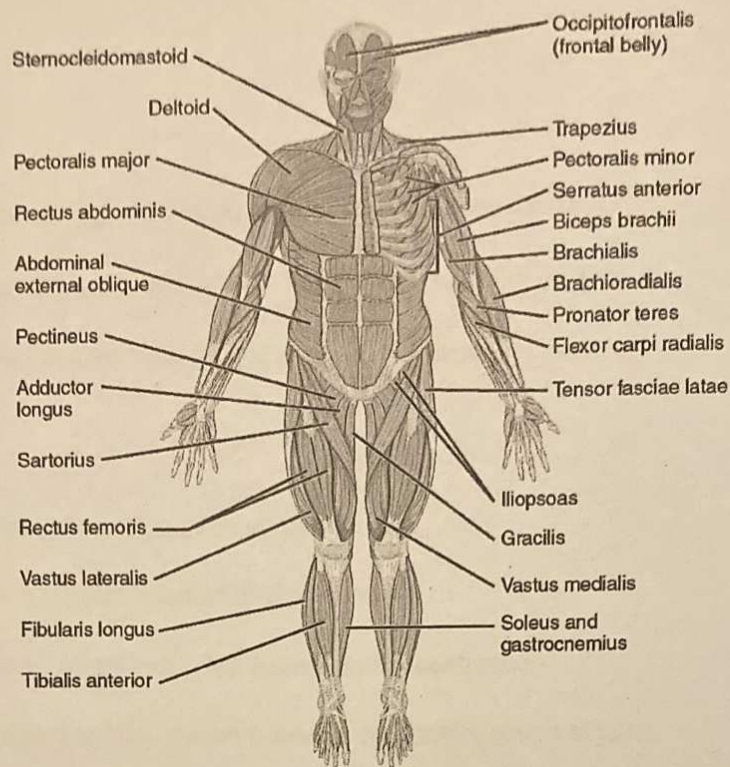
Basic stuff

LARGEST ORGAN IN THE BODY EPIDERMIS REPLACES ITSELF EVERY 2-4 WEEKS
THERMOREGULATION → SWEATING (COOLING) AND SHIVERING (WARMING)
PROTECTION → CHEMICAL, PHYSICAL, AND BIOLOGICAL BARRIER

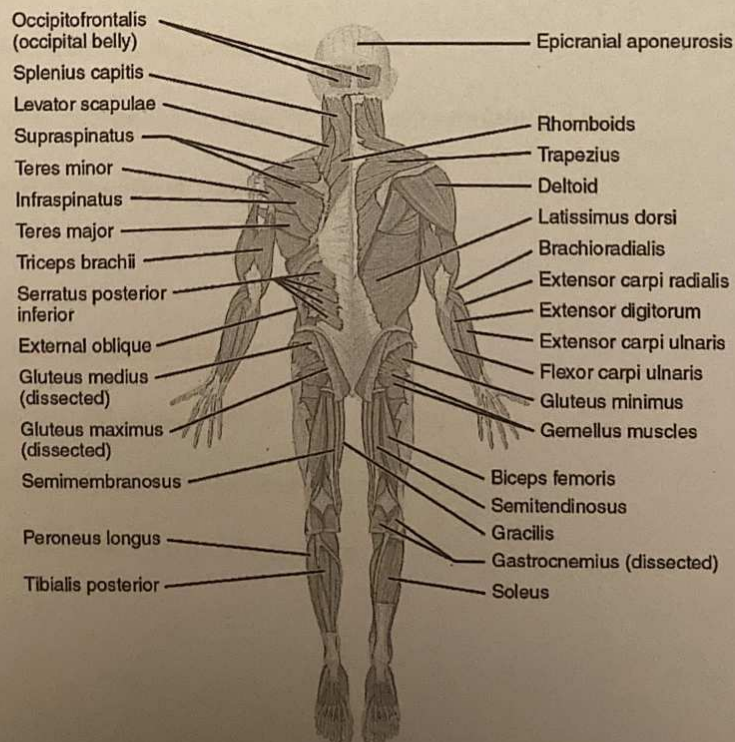
MUSCULAR SYSTEM

General Functions

- Produce movement
 - Maintain posture
 - Generate heat
 - Stabilize joints
 - Support and protect internal structures
-



Major muscles of the body.
Right side: superficial; left side:
deep (anterior view)



Major muscles of the body.
Right side: superficial; left side:
deep (posterior view)

Types

- Skeletal Muscle: Voluntary, striated, attached to bones
 - Smooth Muscle: Involuntary, non-striated, found in organs
 - Cardiac Muscle: Involuntary, striated, found only in the heart
-

Attachment and Function of Skeletal Muscle

- Origin: Fixed attachment; does not move
 - Insertion: Attachment that moves during contraction
 - Function: Muscles pull on bones to create movement at joints
-

Muscle Pairs

- Muscles work in antagonistic pairs
 - One contracts (agonist) while the other relaxes (antagonist)
 - Example: Biceps contracts / Triceps relaxes
-

Contraction of Skeletal Muscles

- Sliding filament mechanism (actin + myosin)
 - Requires ATP, calcium, and nerve impulse
 - Muscle shortens and produces force
-

I. MUSCULAR SYSTEM

A. There are 656 _____ that make up the muscular system.

B. The Muscular System has 3 main responsibilities:

1. Body Movement
2. Body Form & Shape (to Maintain Posture)
3. Body Heat (to Maintain Body Temperature)

C. Muscles are bundles of muscle fibers held together by connective tissue.

D. All muscles have certain properties:

1. Excitability - irritability, the ability to respond to a stimulus such as a nerve impulse
 - a) This is the ability to respond to certain stimuli by producing electrical signs called Action Potentials (impulses).
2. Contractibility - muscle fibers that are stimulated by nerves contract or become short and shrink
3. Extensibility - the ability to be stretched
4. Elasticity - allows the muscle to return to its original shape after it has contracted or stretched

E. There are three main kinds of muscle: cardiac, skeletal, and smooth (visceral).

1. Cardiac muscle - forms the walls of the heart and contracts to circulate blood.

- a) Cells are striated and branched - only found in the heart. Joined in a continuous network without a sheath separation. The membranes of adjacent cells are fused at places called intercalated discs.
- b) A communication system at the fused area will not permit independent cell contraction - when one cell receives a signal to contract, all neighboring cells are stimulated and they contract together to produce a heartbeat.
- c) Cardiac muscle requires a continuous supply of oxygen to function - cut off for as little as 30 seconds, cardiac muscle cells start to die. does not regrow, once it dies it stays dead

2. Smooth (visceral) muscle - found in the internal organs; such as the digestive and respiratory tracts, and blood vessels and eyes. Visceral muscle contracts to cause movements in these organs.

- a) Cardiac and Smooth/Visceral Muscles are involuntary - meaning they function without a conscious thought or control.
- b) Controlled instead by the autonomic nervous system.
- c) Cells are small and small spindle shaped There is only one nucleus located at the center of the cell. They are called smooth muscles because they are unmarked by any distinctive striations.
- d) They are unattached to bones, they act slowly, do not fire easily, and can remain contracted for a long time.

3. Skeletal muscle - attached to bones and causes body movement
- a) Consists of bundles of muscle cells (also known as a muscle fiber) - each cell is multi nucleate (contains many nuclei). The cell membrane is called sarcolemma and the cytoplasm is sarco plasm.
 - b) Skeletal muscle is voluntary - a person has control over its action
 - c) Skeletal muscle has 5 important functions:
 - (1) Attaches to bones to provide voluntary movement
 - (2) Produce heat and energy for the body
 - (3) Help maintain posture by holding the body erect
 - (4) protects internal organs
 - d) Attach to bones in different ways; some attach by tendons (strong, tough, fibrous connective tissue cords), some attach by ~~ligaments~~ Fascia (tough, sheetlike membrane that covers and protects tissue)
 - e) When a muscle attaches to bone, the end that does not move is called the Origin. The end that moves when the muscle contracts is called the insertion.
 - f) The belly is the central body of the muscle.
 - g) Skeletal Muscle Actions/Movements:
 - (1) Adduction - moving a body part toward the midline
 - (2) Abduction - moving a body part away from the midline
 - (3) Flexion - decreasing the angle between two bones, or bending a body part
 - (4) Extension - increasing the angle between two bones, or straightening a body part
 - (5) Rotation - turning a body part around its own axis (turning the head from side to side)
 - (6) Circumduction - moving in a circle at a joint or moving one end of a body part in a circle while the other end remains stationary
 - h) Muscles are partially contracted all the time - the state of partial contraction is called muscle tone - can be achieved through proper nutrition and regular exercise.
 - i) Muscle contractions may be isotonic or isometric - when muscles contract & shorten, it is called an isotonic contraction. When the muscle Tension increases but the muscle does not shorten, it is called an isometric contraction.
 - j) Loss of muscle tone can occur during severe illness or paralysis; when muscles are not used for long periods of time, they can shrink and lose size and strength - this is called atrophy. Lack of use can result in Contracture - a severe tightening of a flexor muscle resulting in a bending of a joint. The opposite is if from over exercise, muscles will become enlarged - hypertrophy

4. Other types of muscle:

- a) Sphincter (dilator) muscles - special circular muscles in the openings between the esophagus and stomach, and the stomach and small intestine; also found in the walls of the anus, the urethra, and the mouth. They open and close to control the passage of substances.

F. Muscles of the body are arranged in pairs.

1. prime mover - produces movement in a single direction
2. Antagonist - produces movement in the opposite direction
^ Antagonist Pair
3. Synergists - group of muscles that help to steady a movement or stabilize joint activity.

G. For muscles to contract and do their work, they need energy. The major source of this energy is ATP (adenosine triphosphate), a compound found in muscle cells. To make ATP, the cell requires oxygen, glucose, and other material that is brought to the cell by circulating blood.

H. Movement of muscles occurs as a result of 2 major events:

1. Myoneural Stimulation
2. Contraction of muscle Proteins

Skeletal muscles must be stimulated by nerve impulses to contract.

A single muscle contraction is called a Twitch.

A motor neuron stimulates all of the skeletal muscles within a motor unit (a motor neuron plus all the muscle fibers it stimulates).

The junction between the motor neuron's fiber (axon), which transmits the impulse, and the muscle cell's sarcolemma (muscle cell membrane) is the Neuromuscular Junction.

The gap between the axon and the end of the muscle cell is known as the Synaptic Cleft.

When the nerve impulses reach the end of the axon, acetylcholine (neurotransmitter) is released. Diffuses across the synaptic cleft and attaches to receptors on the sarcolemma.

The sarcolemma then is temporarily permeable to sodium ions (Na^+) which rushes into the muscle cell. This gives an imbalance in the muscle cell (excessive positive ions), and changes the electrical conduction of the sarcolemma.

This causes the Action Potential.

Skeletal muscle contraction begins with action potential, traveling over the entire surface of the sarcolemma conducting the electric impulse from one end of the cell to the other. This results in the contraction of the muscle cell. This causes calcium ions (Ca^{++}) to be released from storage areas in the muscle cells).

When calcium ions attach to the action myofilaments, the cell shortens, and the sliding of the myofilaments is energized by ATP.

I. Returning to Resting Phase

1. Diffusion of potassium and sodium ions back to their initial positions outside the cell.
2. When action potential ends, calcium ions are reabsorbed into their storage areas and the muscle cell relaxes and returns to its original length.
3. The strength of a contraction depends on:
 - a) The strength of the stimulus (a weak stimulus will not bring about a contraction).
 - b) The duration of the stimulus (even if the stimulus is strong when only applied for a millisecond, it may not be long enough to be effective), the speed of the application (strong stimulus applied quickly and quickly pulled away may not have enough time to take effect).

4. ALL OR NONE LAW:

01011 A muscle cell, when stimulated properly, contracts all the way.
The muscle cell relaxes until it is stimulated by the next release of acetylcholine.

- J. Muscle Fatigue - caused by an accumulation of lactic acid in the muscle. During periods of vigorous exercise, the blood is unable to transport enough oxygen for the complete oxidation of glucose in the muscles - causing the muscles to contract anaerobically (without oxygen).

Lactic acid normally leaves muscle, passing into the bloodstream; but if vigorous exercise continues, the lactic acid level in the bloodstream rises sharply. Lactic acid accumulates within the muscle.

This impedes muscular contraction, causing muscle fatigue & cramps. It can lead to further illness, injury if not corrected quickly enough.

K. Principle Skeletal Muscles

1. Naming of Skeletal Muscles

- a) Named by location, size, direction, number of origins, location of origin and insertion, and action (most)

Fatigue

- Muscle cannot contract efficiently
 - Caused by low oxygen, low ATP, waste buildup (lactic acid)
-

Tone

- Continuous partial contraction
 - Keeps body upright
 - Maintains readiness for movement
-

Muscles of Head and Neck

- Frontalis: Raises eyebrows
 - Temporalis: Helps chew
 - Masseter: Closes jaw
 - Orbicularis oculi: Closes eyelids
 - Orbicularis oris: Controls mouth/lips
 - Sternocleidomastoid: Turns and flexes head
 - Trapezius: Moves shoulders and extends neck
-

Upper Extremities

- Deltoid: Abducts arm
 - Biceps brachii: Flexes elbow
 - Triceps brachii: Extends elbow
 - Brachioradialis: Assists elbow flexion
 - Rotator cuff (group): Stabilizes shoulder joint
-

Trunk

- Pectoralis major: Moves arm across body
 - Intercostals: Assist breathing
 - Rectus abdominis: Flexes trunk
 - Obliques: Rotate and bend trunk
 - Latissimus dorsi: Extends and adducts arm
 - Erector spinae: Extends spine
-

Lower Extremity

- Gluteus maximus: Extends thigh
- Gluteus medius: Abducts thigh
- Quadriceps group: Extends knee
- Hamstrings: Flex knee, extend hip
- Gastrocnemius: Plantar flexion

- Soleus: Assists plantar flexion
-

Arms / Wrist / Hand

- Flexor carpi group: Flex wrist
 - Extensor carpi group: Extend wrist
 - Flexor digitorum: Flex fingers
 - Extensor digitorum: Extend fingers
 - Intrinsic hand muscles: Control fine motor movements
-

Leg / Ankle / Foot / Toes

- Tibialis anterior: Dorsiflexes foot
 - Fibularis (peroneal) group: Everts foot
 - Extensor digitorum longus: Extends toes
 - Flexor digitorum longus: Flexes toes
 - Gastrocnemius/Soleus: Plantar flex foot
 - Intrinsic foot muscles: Maintain arch, toe control
-

DISORDERS

Strain –

Symptoms: Muscle tear, pain, swelling.

Treatment: Rest, ice, compression.

Spasm –

Symptoms: Sudden involuntary muscle contraction.

Treatment: Stretching, hydration.

Myalgia –

Symptoms: General muscle pain.

Treatment: Rest, NSAIDs.

Hernia –

Symptoms: Bulging tissue through muscle wall.

Treatment: Surgery, support belt.

Flat Feet –

Symptoms: Low arches, foot pain.

Treatment: Orthotics, supportive shoes.

Tetanus –

Symptoms: Muscle stiffness, lockjaw.

Treatment: Antitoxin, vaccine, antibiotics.

Torticollis –

Symptoms: Twisted neck, limited movement.

Treatment: Physical therapy, heat, stretching.

Dystrophy (Muscular Dystrophy) –

Symptoms: Progressive muscle weakening.

Treatment: Supportive therapy, mobility assistance.

Myasthenia Gravis –

Symptoms: Muscle weakness, drooping eyelids.

Treatment: Anticholinesterase meds, immunotherapy.

Tennis Elbow –

Symptoms: Pain in lateral elbow.

Treatment: Rest, brace, NSAIDs.

Shin Splints –

Symptoms: Pain along shin bone.

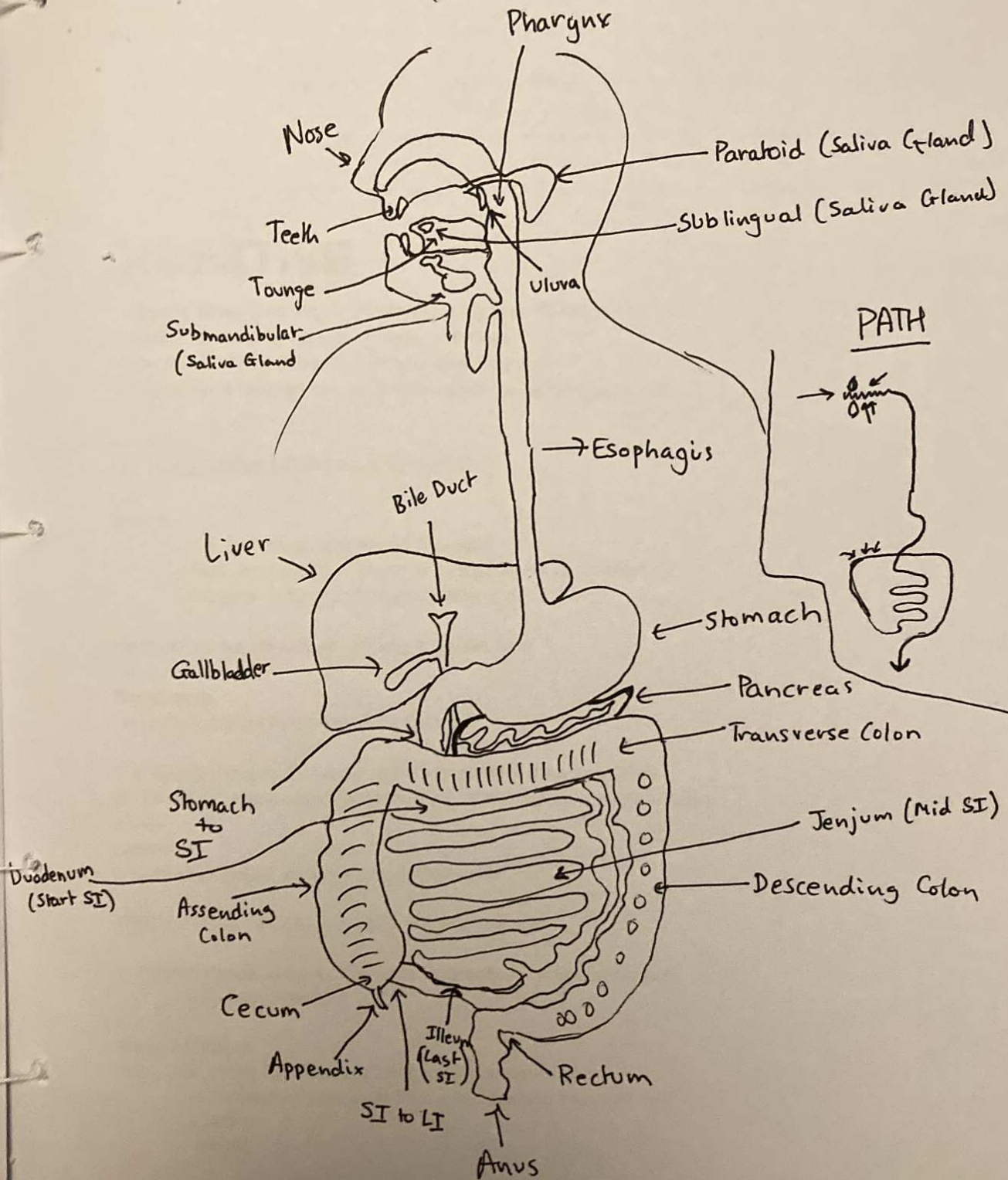
Treatment: Rest, ice, proper footwear.

Rotator Cuff Disease –

Symptoms: Shoulder pain, weakness when lifting.

Treatment: Physical therapy, anti-inflammatory meds, surgery (severe).

DIGESTIVE



DIGESTIVE

- Breaks down food into nutrients the body and absorb
- Nutrients used for growth, energy and repair
- Two types of digestion, mechanical and chemical
- Food travels through one long tube called the alimentary canal

ALL FOLLOWING STEPS ARE IN ORDER

Mouth

- Teeth -> help mechanical digestion
- Salivary Glands -> produce saliva that contains amylase
- Tongue -> Mix saliva with chewed food

Mechanical digestion starts as mouth chews food

Esophagus

Muscular tube connects mouth to stomach

Transports using rhythmic muscle contractions called peristalsis
No chemical digestion or mechanical digestion happens only transport

Stomach

J shaped muscular sac with thick folded walls

MECHANICAL DIGESTION with churning motions

- Gastric glands make gastric juice (hydrochloric acid and pepsin)
- Acid kills bacteria

Small Intestine

Long Thin coiled tube that is 20 ft long w/ 3 sections

- Duodenum - gets juice from pancreas and bile from liver
- Jejunum
- Ileum

Walls have finger-like projections called villi to increase surface area
Nutrients pass from walls into bloodstream

PRIMARY SITE OF CHEMICAL DIGESTION

Pancreas

Large gland behind the stomach

Produces enzymes that digest carbs, protein and fat
Secretes bicarbonate to neutralize stomach acid that gets into the small intestine
Makes proper pH

Liver

Largest Internal Organ, upper right abdomen

Produces Bile which breaks down fats into smaller drops
Processes nutrients after the Small intestine
Detox for harmful substances
Stores vitamins

Gallbladder

Small pear shaped sac below liver

- Stores and concentrates bile
- Releases bile into intestine with fatty food

Large Intestine

Wide Tube 5ft

- Absorbs water and electrolytes from digested food
- Houses bacteria that make vitamins
- Compacts waste into poop
- Stores poop till

Rectum & Anus

Rectum stores feces temporarily
Anus where it leaves
